Reading Assignment: Tracking Aviation's Footprint from the Sky

Critical review of *"Global Civil Aviation Emissions Estimates for 2017–2020 Using ADS-B Data"* (Quadros, Snellen, Sun & Dedoussi, *Journal of Aircraft*, 2022)

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1. Problem

Aviation is responsible for roughly 2–3% of global anthropogenic CO_2 emissions, and its total climate effect is larger once non-CO_2 effects such as NO_x and contrails are included [3]. Despite this, our knowledge of *where* and *when* aircraft emit is surprisingly limited. Official figures (ICAO, IEA) are mostly *top-down*: they divide jet-fuel sales between countries, come out with a delay of a year or more, and say nothing about the altitude or location of emissions, which atmospheric models need. Earlier *bottom-up* inventories, built from flight schedules or radar data, are either proprietary, regionally limited, or not updated every year. This became a clear problem in 2020: when COVID-19 grounded much of the world's fleet, there was no timely, open and spatially resolved estimate of how much aviation emissions actually fell. The question the authors ask is whether the huge volume of aircraft position data now broadcast via ADS-B (Automatic Dependent Surveillance–Broadcast) can be turned into a reliable, up-to-date global emissions inventory [1].

2. Approach

The authors build a bottom-up pipeline that turns raw flight observations into gridded, speciated emissions:

- **Activity data.** Flights observed by crowdsourced networks of ground ADS-B receivers are aggregated into flight lists of (aircraft type, origin, destination, count) for 2017–2020.
- **Trajectory model.** Each origin–destination pair is flown along the great circle, extended by a constant lateral-inefficiency term (40.5 NM + 3.87% of the distance), with cruise levels set from observed data and monthly-averaged winds.
- **Fuel burn.** Climb, cruise and descent are simulated with EUROCONTROL's BADA 3.15 aircraft performance model; the landing–takeoff cycle uses a time-in-mode approach.
- **Emissions.** CO_2, H_2O and SO_x scale with fuel; NO_x, CO and HC use ICAO Engine Emissions Databank indices corrected for altitude (Boeing Fuel Flow Method 2); non-volatile particulate matter (nvPM) is estimated from engine smoke numbers.
- **Openness.** The model is released as open-source software (openAVEM) and the gridded monthly results as a CC BY 4.0 dataset.

Key results. For 2019 the model estimates about 297 Tg of fuel, 937 Tg of CO_2 and 4.62 Tg of NO_x. COVID-19 cut 2020 fuel burn by about 48% relative to expected levels, avoiding roughly 463 Tg of CO_2 and 2.29 Tg of NO_x. The authors also track changes in fleet-average emission indices over time and conclude that ADS-B is a viable source for *timely and transparent* global emissions monitoring.

3. Critical Reflection

Where we agree. The central idea is convincing and, in our view, important. The paper shows that a data stream originally built for air-traffic safety can be reused as an environmental sensor. Producing a global inventory within months instead of years, with full 3-D resolution, is a real step forward: it made it possible to quantify the COVID-19 "natural experiment" almost in real time. Its estimates also hold up well against independent work: the later GAIA inventory, which uses full satellite and terrestrial ADS-B trajectories, finds 2019 fuel burn only 4.7% lower and an almost identical 2020 drop (−48.2% vs. −47.1%) [2]. Releasing the code and gridded data is good scientific practice and makes the work useful to climate modellers.

Where we are more critical.

1. *"Transparent" is only partly true.* The pipeline depends on BADA 3, which must be licensed from EUROCONTROL, and on flight lists derived from a commercial tracking platform [2]. An outside

group cannot fully reproduce the inventory from public inputs. A truly open version would need open performance models (e.g. OpenAP) and open data (e.g. the OpenSky Network [4]).

2. *Coverage bias.* Ground receivers are dense in Europe and North America but sparse over oceans, Africa and parts of Asia. Missing flights simply do not appear, which helps explain why the result (297 Tg) stays below the IEA's top-down 320 Tg [2]. This bias is not neutral: it under-represents exactly the regions where air traffic is growing fastest.

3. *The data is richer than the model.* ADS-B provides real trajectories, yet the method reduces them to origin–destination counts and re-flies them as idealised great circles with monthly winds. This throws away actual routing, holding patterns, airspace closures and conflict-zone detours; GAIA shows that a fixed 4–6% inefficiency underestimates distance flown in regions like Europe [2].

4. *Emissions are not climate impact.* The paper reports emitted mass, but contrail cirrus may cause a larger warming effect than aviation's CO_2 itself [3]. Location-aware ADS-B data is exactly what is needed to assess contrails, so stopping at emissions leaves the most policy-relevant question open. nvPM, which drives contrail formation, is also likely underestimated [2].

Overall position. We agree with the paper's main conclusion: ADS-B data *can* support timely, global, spatially resolved emissions monitoring, and the COVID-19 results are robust. We disagree with how strongly it presents the approach as transparent, and we think its simplified trajectories under-use the very data that makes it novel. Our own project, which ingests OpenSky state vectors into a Spark/MongoDB pipeline to detect holding patterns and congestion, illustrates the next step: using real trajectories, not idealised ones, to attribute emissions to specific inefficiencies that operational changes could actually remove.

References

[1] F. D. A. Quadros, M. Snellen, J. Sun, I. C. Dedoussi, "Global civil aviation emissions estimates for 2017–2020 using ADS-B data," *Journal of Aircraft*, 59(6):1394–1405, 2022. doi:10.2514/1.C036763.

[2] R. Teoh, Z. Engberg, M. Shapiro, L. Dray, M. E. J. Stettler, "The high-resolution Global Aviation emissions Inventory based on ADS-B (GAIA) for 2019–2021," *Atmos. Chem. Phys.*, 24:725–744, 2024.

[3] D. S. Lee et al., "The contribution of global aviation to anthropogenic climate forcing for 2000 to 2018," *Atmospheric Environment*, 244:117834, 2021.

[4] M. Schäfer, M. Strohmeier, V. Lenders, I. Martinovic, M. Wilhelm, "Bringing up OpenSky: A large-scale ADS-B sensor network for research," *Proc. IPSN*, 2014.